



Structural Design Report

180' Monopole

Site: US-NY-2029, NY

Prepared for: PTI SERVICES

by: Sabre IndustriesTM

Job Number: 25-3277-JDS-R1

April 21, 2025

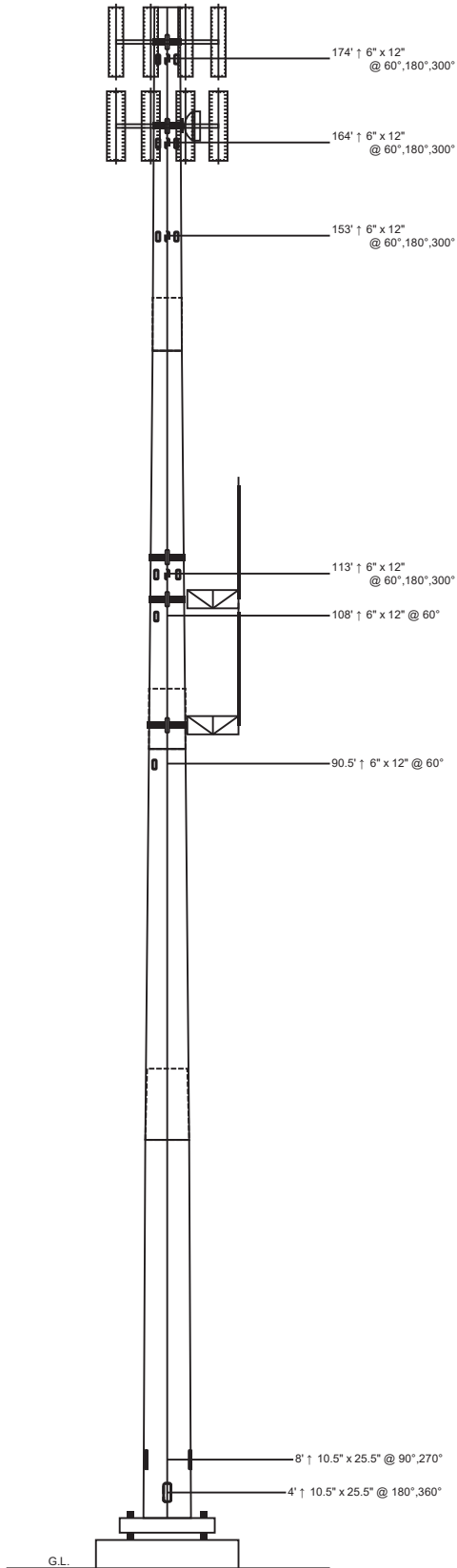
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Foundation Calculations.....	25-35



DATE OF RENEWAL: 7/31/23

DATE OF EXPIRATION: 7/31/26

Length (ft)	53'-3"	53'-6"	53'-6"	40'-9"
Number Of Sides	18			
Thickness (in)	3/8"	8" - 6"	7" - 3"	5/16"
Lap Splice (ft)				6" - 3"
Top Diameter (in)	58.21"	50.29"	42.02"	36"
Bottom Diameter (in)	68.46"	60.59"	52.32"	43.84"
Taper (in/ft)				
Grade				
Weight (lbs)	16267	12697	9106	6041
Overall Steel Height (ft)				



Design Criteria - ANSI/TIA-222-H

Wind Speed (No Ice)	109 mph
Wind Speed (Ice)	40 mph
Design Ice Thickness	1.50 in
Risk Category	II
Exposure Category	C
Topographic Factor Procedure	Method 1 (Simplified)
Topographic Category	1
Ground Elevation	435 ft
Seismic Importance Factor, I _e	1.00
0.2-sec Spectral Response, S _s	0.142 g
1-sec Spectral Response, S ₁	0.051 g
Site Class	D (DEFAULT)
Seismic Design Category	B
Basic Seismic Force-Resisting System	Telecommunication Tower (Pole: Steel)

Limit State Load Combination Reactions

Load Combination	Axial (kips)	Shear (kips)	Moment (ft-k)	Deflection (ft)	Sway (deg)
1.2 D + 1.0 W _o	70.19	48.56	6428.91	9.67	5.06
0.9 D + 1.0 W _o	52.76	48.33	6336.84	9.49	4.96
1.2 D + 1.0 D _i + 1.0 W _i	117.2	11.88	1634.92	2.49	1.3
1.2 D + 1.0 E _v + 1.0 E _h	71.69	1.73	258.43	0.4	0.21
0.9 D - 1.0 E _v + 1.0 E _h	50.69	1.72	254.23	0.4	0.21
1.0 D + 1.0 W _o (Service @ 60 mph)	58.21	13.22	1739.79	2.61	1.36

Base Plate Dimensions

Shape	Diameter	Thickness	Bolt Circle	Bolt Qty	Bolt Diameter
Round	81.25"	2"	75.5"	18	2.25"

Anchor Bolt Dimensions

Length	Diameter	Hole Diameter	Weight	Type	Finish
84"	2.25"	2.625"	2179.8	A615-75	Galv

Notes

- 1) Antenna Feed Lines Run Inside Pole
- 2) All dimensions are above ground level, unless otherwise specified.
- 3) Weights shown are estimates. Final weights may vary.
- 4) This tower design and, if applicable, the foundation design(s) shown on the following page(s) also meet or exceed the requirements of the 2018 International Building Code.
- 5) Full Height Step Bolts
- 6) Tower Rating: 98.7%



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Job:	25-3277-JDS-R1
Customer:	PTI SERVICES
Site Name:	US-NY-2029, NY
Description:	180' Monopole
Date:	4/21/2025
By:	REB

Designed Appurtenance Loading

Elev	Description	Tx-Line
176	SitePro1 F4P-12W	
176	(8) NNH4-65C-R6	(9) 1"
176	(4) AIR 6472 B77G B77M	
176	(4) RRU 4890	
176	(4) RRU 4490	
176	(4) RRU (17" x 18" x 7")	
176	(3) DC9-48-60-24-8C-EV	(3) 1/2"
166	(1) Dish Mount (Monopole Only) - Pipe Mount (up to 6" Dish)	
166	SitePro1 F4P-12W	
166	(4) AIR 6419 B41	(4) 6x24 Hybrid
166	(4) RRUS 4460 B25+B66	(2) 1 1/2"
166	(4) RRUS 4480 B71+B85	
166	(2) IP-20C	(2) 1/4"
166	(4) 840590966	

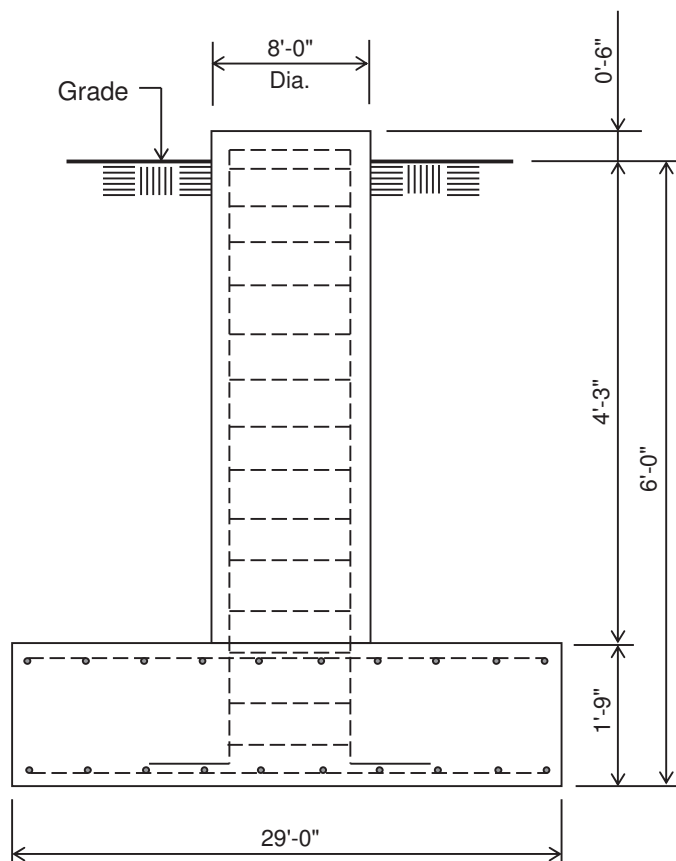
Elev	Description	Tx-Line
166	(1) VHLP3-11W	
155	(1) 250 Sq. Ft. EPA (8000 lbs)	(6) 1 5/8"
116.72	(1) SC46A-HF1LDF(DXX-PIP)	(1) 7/8"
115	(3) Armor Tower Arched Boom - 8' Face	
115	(6) NHH-55C-R2B	
115	(1) RVZDC-6627-PF-48	
115	(1) RxxDC-3315-PF-48	(3) 1-5/8 Hybrid
115	(3) RF4461d-13a	
115	(3) RF4801d-25B	
115	(3) MT6413-77A	
110	6ft Sidearm	
110	(1) TTA (31.25" x 13.25" x 11.5")	(1) 1/2"
101.72	(1) SC46A-HF1LDF(DXX-PIP)	(1) 7/8"
95	6ft Sidearm	

 Sabre Industries 7101 Southbridge Drive P.O. Box 658 Sioux City, IA 51102-0658 Phone: (712) 258-6690 Fax: (712) 279-0614 Information contained herein is the sole property of Sabre Communications Corporation, constitutes a trade secret as defined by Iowa Code Ch. 550 and shall not be reproduced, copied or used in whole or part for any purpose whatsoever without the prior written consent of Sabre Communications Corporation.	Job: 25-3277-JDS-R1	
	Customer: PTI SERVICES	
	Site Name: US-NY-2029, NY	
	Description: 180' Monopole	
	Date: 4/21/2025	By: REB

Customer: PTI SERVICES

Site: US-NY-2029, NY

180' Monopole



ELEVATION VIEW

(63.35 Cu. Yds.)

(1 REQUIRED; NOT TO SCALE)

Notes:

- 1) Concrete shall have a minimum 28-day compressive strength of 4,500 psi, in accordance with ACI 318-14.
- 2) Rebar to conform to ASTM specification A615 Grade 60.
- 3) All rebar to have a minimum of 3" concrete cover.
- 4) All exposed concrete corners to be chamfered 3/4".
- 5) The foundation design is based on the geotechnical report by Delta Oaks Group, Project No. GEO24-23357-08, dated November 17, 2024.
- 6) See the geotechnical report for compaction requirements, if specified.
- 7) 4.25 ft of soil cover is required over the entire area of the foundation slab.
- 8) The bottom anchor bolt template shall be positioned as closely as possible to the bottom of the anchor bolts.

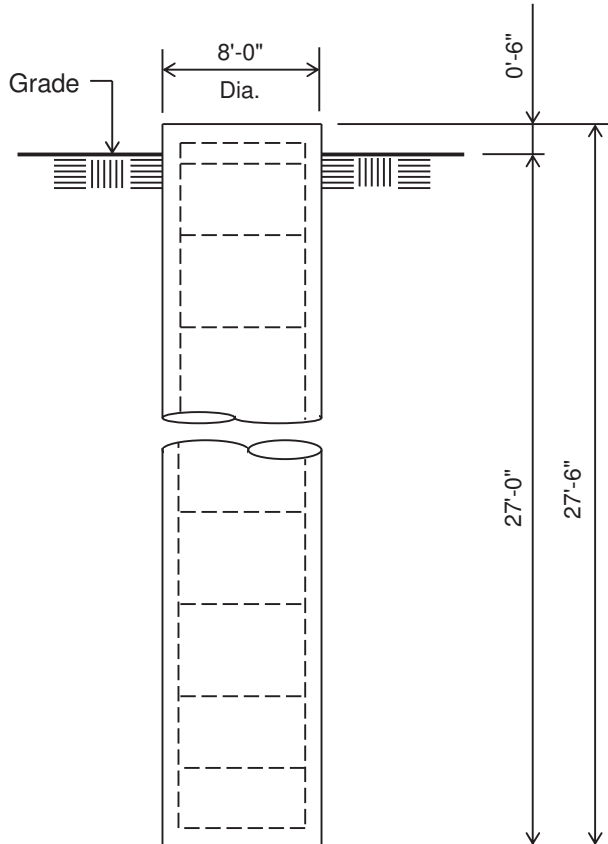
Rebar Schedule for Pad and Pier

Pier	(50) #8 vertical rebar w/ hooks at bottom w/ #5 ties, (2) within top 5" of pier, then 4" C/C
Pad	(51) #10 horizontal rebar evenly spaced each way top and bottom (204 total)

Customer: PTI SERVICES

Site: US-NY-2029, NY

180' Monopole



ELEVATION VIEW

(51.20 Cu. Yds.)

(1 REQUIRED; NOT TO SCALE)

Notes:

- 1) Concrete shall have a minimum 28-day compressive strength of 4,500 psi, in accordance with ACI 318-14.
- 2) Rebar to conform to ASTM specification A615 Grade 60.
- 3) All rebar to have a minimum of 3" concrete cover.
- 4) All exposed concrete corners to be chamfered 3/4".
- 5) The foundation design is based on the geotechnical report by Delta Oaks Group, Project No. GEO24-23357-08, dated November 17, 2024.
- 6) See the geotechnical report for drilled pier installation requirements, if specified.
- 7) The bottom anchor bolt template shall be positioned as closely as possible to the bottom of the anchor bolts.

Rebar Schedule for Pier

Pier	(52) #8 vertical rebar w/ #5 ties, (2) within top 5" of pier, then 7" C/C
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(USA 222-H) - Monopole Spatial Analysis (c)2017 Guymast Inc.

Tel: (416) 736-7453 Fax: (416) 736-4372 Web: www.guymast.com

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180' Monopole / US-NY-2029, NY

* All pole diameters shown on the following pages are across corners.
See profile drawing for widths across flats.

POLE GEOMETRY

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ELEV	SECTION	No.	OUTSIDE	THICK	RESISTANCES	SPLICE	...OVERLAP...	w/t
ft	NAME	SIDE	DIAM	-NESS	*Pn *Mn	TYPE	LENGTH RATIO	
			in	in	kip ft-kip		ft	
179.0								
	A	18	36.56	0.312	2508.9 1840.0			19.3
			43.29	0.312	2808.8 2446.1			
144.5			43.29	0.312	2808.8 2446.1			
	A/B	18	43.90	0.312	2833.3 2502.5	SLIP	6.25 1.73	
			43.90	0.312	2833.3 2502.5			
138.2								
	B	18	51.69	0.312	3110.1 3242.1			23.3
			51.69	0.312	3110.1 3242.1			
98.2								
	B/C	18	52.50	0.375	4063.8 4292.4	SLIP	7.25 1.68	
			52.50	0.375	4063.8 4292.4			
91.0								
	C	18	59.85	0.375	4386.1 5291.6			23.4
			59.85	0.375	4386.1 5291.6			
53.2								
	C/D	18	60.78	0.375	4422.1 5418.4	SLIP	8.50 1.70	
			60.78	0.375	4422.1 5418.4			
44.7								
	D	18	69.51	0.375	4714.7 6617.7			27.2
0.0								

POLE ASSEMBLY

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SECTION	BASE	BOLTS AT BASE OF SECTION			CALC
NAME	ELEV	NUMBER	TYPE	DIAM	BASE
	ft			in	ELEV
					ft
A	138.250	0	A325	0.00	138.250
B	91.000	0	A325	0.00	91.000
C	44.750	0	A325	0.00	44.750
D	0.000	0	A325	0.00	0.000

POLE SECTIONS

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SECTION	No. of	LENGTH	OUTSIDE	DIAMETER	BEND	MAT-	FLANGE	FLANGE	FLANGE
NAME	SIDES		BOT	TOP	RAD	ERIAL	BOT	TOP	WELD
			*	*		ID			GROUP
		ft	in	in	in				ID
A	18	40.75	44.52	36.56	0.625	1	0	0	0
B	18	53.50	53.12	42.66	0.625	2	0	0	0
C	18	53.50	61.53	51.07	0.625	3	0	0	0
D	18	53.25	69.51	59.10	0.625	4	0	0	0

* - Diameter of circumscribed circle

MATERIAL TYPES =====

TYPE OF SHAPE	TYPE NO	NO OF ELEM.	ORIENT	HEIGHT	WIDTH	.THICKNESS.		IRREGULARITY	
						WEB	FLANGE	.PROJECTION. % OF AREA	ORIENT
			& deg	in	in	in	in		deg
PL	1	1	0.0	44.52	0.31	0.312	0.312	0.00	0.0
PL	2	1	0.0	53.12	0.31	0.312	0.312	0.00	0.0
PL	3	1	0.0	61.53	0.38	0.375	0.375	0.00	0.0
PL	4	1	0.0	69.51	0.38	0.375	0.375	0.00	0.0

& - With respect to vertical

MATERIAL PROPERTIES =====

MATERIAL TYPE NO.	ELASTIC MODULUS ksi	UNIT WEIGHT pcf	.. STRENGTH ..		THERMAL COEFFICIENT /deg
			Fu ksi	Fy ksi	
1	29000.0	490.0	80.0	65.0	0.00001170
2	29000.0	490.0	80.0	65.0	0.00001170
3	29000.0	490.0	80.0	65.0	0.00001170
4	29000.0	490.0	80.0	65.0	0.00001170

* Only 5 condition(s) shown in full

* RRUs/TMAs were assumed to be behind antennas

* Some concentrated wind loads may have been derived from full-scale wind tunnel testing

LOADING CONDITION A

109 mph wind with no ice. Wind Azimuth: 0° (1.2 D + 1.0 Wo)

LOADS ON POLE =====

LOAD TYPE	ELEV ft	APPLY. RADIUS ft	LOAD..AT AZI	LOAD AZI	FORCES.....		MOMENTS.....	
					HORIZ kip	DOWN kip	VERTICAL ft-kip	TORSNAL ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.2726	0.0000	0.0000
C	175.000	0.00	0.0	0.0	6.2439	2.4118	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0293	0.0151	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.6909	0.0000	0.0000
C	165.000	0.00	0.0	0.0	5.6195	1.3299	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0321	0.0168	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0317	0.0168	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	1.1532	0.0000	0.0000
C	154.000	0.00	0.0	0.0	10.8563	9.6000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0313	0.0168	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0308	0.0168	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0303	0.0168	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0750	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.1731	0.0342	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0298	0.0168	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.7387	0.0000	0.0000
C	114.000	0.00	0.0	0.0	4.6962	1.8970	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0523	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1394	0.0600	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.5037	0.8488	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0292	0.0168	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0653	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.1682	0.0342	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0286	0.0168	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.4884	0.8488	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0279	0.0168	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0168	0.0000	0.0000

C	65.000	0.00	0.0	0.0	0.0264	0.0168	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0255	0.0168	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0244	0.0168	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0232	0.0168	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0216	0.0168	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0194	0.0168	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0870	0.1466	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0968	0.1695	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0983	0.3453	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0983	0.3453	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0988	0.1761	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1070	0.2027	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1079	0.4539	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.4539	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.2520	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1100	0.2771	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1096	0.2821	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1096	0.2821	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1091	0.5738	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1091	0.5738	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1082	0.2917	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0946	0.3178	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0948	0.3230	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0963	0.3282	0.0000	0.0000

ANTENNA LOADING

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.....ANTENNA.....	ATTACHMENT		ANTENNA FORCES.....					
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.47	0.00	0.14	0.00

=====

LOADING CONDITION M

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109 mph wind with no ice. Wind Azimuth: 0° (0.9 D + 1.0 Wo)

LOADS ON POLE

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LOAD	ELEV	APPLY..LOAD..AT	LOAD	FORCES.....		MOMENTS.....		
TYPE	ft	RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL	TORSNAL
		ft			kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0000	0.9544	0.0000	0.0000
C	175.000	0.00	0.0	0.0	6.2439	1.8088	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0293	0.0113	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.2682	0.0000	0.0000
C	165.000	0.00	0.0	0.0	5.6195	0.9974	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0321	0.0126	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0317	0.0126	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	0.8649	0.0000	0.0000
C	154.000	0.00	0.0	0.0	10.8563	7.2000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0313	0.0126	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0308	0.0126	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0303	0.0126	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0562	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.1731	0.0257	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0298	0.0126	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.5540	0.0000	0.0000
C	114.000	0.00	0.0	0.0	4.6962	1.4227	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0392	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1394	0.0450	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.5037	0.6366	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0292	0.0126	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0489	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.1682	0.0257	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0286	0.0126	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.4884	0.6366	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0279	0.0126	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0126	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0264	0.0126	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0255	0.0126	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0244	0.0126	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0232	0.0126	0.0000	0.0000

C	25.000	0.00	0.0	0.0	0.0216	0.0126	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0194	0.0126	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0870	0.1099	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0968	0.1271	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0983	0.2589	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0983	0.2589	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0988	0.1321	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1070	0.1520	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1079	0.3404	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.3404	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.1890	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1100	0.2078	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1096	0.2116	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1096	0.2116	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1091	0.4303	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1091	0.4303	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1082	0.2188	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0946	0.2383	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0948	0.2422	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0963	0.2461	0.0000	0.0000

ANTENNA LOADING

=====

.....ANTENNA.....			ATTACHMENT		ANTENNA FORCES.....			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.47	0.00	0.10	0.00

=====

LOADING CONDITION Y

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40 mph wind with 1.5 ice. Wind Azimuth: 0* (1.2 D + 1.0 Di + 1.0 Wi)

LOADS ON POLE

=====

LOAD TYPE	ELEV ft	APPLY.. RADIUS ft	LOAD..AT AZI	LOAD AZI	FORCES.....		MOMENTS.....	
					HORIZ kip	DOWN kip	VERTICAL ft-kip	TORSNAL ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.2726	0.0000	0.0000
C	175.000	0.00	0.0	0.0	1.4926	6.9462	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0315	0.0271	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.6909	0.0000	0.0000
C	165.000	0.00	0.0	0.0	1.2638	3.8298	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0344	0.0288	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0338	0.0288	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	1.1532	0.0000	0.0000
C	154.000	0.00	0.0	0.0	2.4860	23.6076	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0331	0.0288	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0324	0.0288	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0317	0.0288	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0750	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0465	0.0827	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0309	0.0288	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.7387	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.9855	7.6184	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0523	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0256	0.1683	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1307	1.1871	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0301	0.0288	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0653	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0448	0.0820	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0292	0.0288	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.1257	1.1822	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0282	0.0288	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0288	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0261	0.0288	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0248	0.0288	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0234	0.0288	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0217	0.0288	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0196	0.0288	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0169	0.0288	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0224	0.2308	0.0000	0.0000

D	144.500	0.00	180.0	0.0	0.0246	0.2644	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0249	0.4421	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0249	0.4421	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0250	0.2737	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0268	0.3112	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0270	0.5641	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0270	0.5641	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0269	0.3628	0.0000	0.0000
D	64.036	0.00	180.0	0.0	0.0273	0.3880	0.0000	0.0000
D	64.036	0.00	180.0	0.0	0.0272	0.3941	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0271	0.4000	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0270	0.6927	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0270	0.6927	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0267	0.4109	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0232	0.4341	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0232	0.4348	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0234	0.4318	0.0000	0.0000

ANTENNA LOADING

=====

.....ANTENNA.....	ATTACHMENT		ANTENNA FORCES.....					
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.08	0.00	0.51	0.00

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LOADING CONDITION AK

Seismic - Azimuth: 0° (1.2 D + 1.0 Ev + 1.0 Eh)

LOADS ON POLE

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LOAD	ELEV	APPLY..LOAD..AT	LOAD	FORCES.....		MOMENTS.....	
TYPE	ft	RADIUS	AZI	HORIZ	DOWN	VERTICAL	TORSIONAL
		ft		kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0751	1.3046	0.0000
C	175.000	0.00	0.0	0.0	0.1423	2.4725	0.0000
C	174.500	0.00	0.0	0.0	0.0009	0.0155	0.0000
C	165.000	0.00	0.0	0.0	0.0887	1.7335	0.0000
C	165.000	0.00	0.0	0.0	0.0186	0.3629	0.0000
C	165.000	0.00	0.0	0.0	0.0697	1.3635	0.0000
C	165.000	0.00	0.0	0.0	0.0009	0.0172	0.0000
C	158.620	0.00	0.0	0.0	0.3160	6.6850	0.0000
C	155.000	0.00	0.0	0.0	0.0008	0.0172	0.0000
C	154.000	0.00	0.0	0.0	0.4386	9.8416	0.0000
C	154.000	0.00	0.0	0.0	0.0527	1.1822	0.0000
C	145.000	0.00	0.0	0.0	0.0007	0.0172	0.0000
C	135.000	0.00	0.0	0.0	0.0006	0.0172	0.0000
C	125.000	0.00	0.0	0.0	0.0005	0.0172	0.0000
C	117.750	0.00	0.0	0.0	0.2705	10.3817	0.0000
C	115.720	0.00	0.0	0.0	0.0009	0.0351	0.0000
C	115.720	0.00	0.0	0.0	0.0019	0.0769	0.0000
C	115.000	0.00	0.0	0.0	0.0004	0.0172	0.0000
C	114.000	0.00	0.0	0.0	0.0475	1.9447	0.0000
C	114.000	0.00	0.0	0.0	0.0185	0.7573	0.0000
C	109.000	0.00	0.0	0.0	0.0014	0.0615	0.0000
C	109.000	0.00	0.0	0.0	0.0012	0.0536	0.0000
C	109.000	0.00	0.0	0.0	0.0194	0.8702	0.0000
C	105.000	0.00	0.0	0.0	0.0004	0.0172	0.0000
C	100.720	0.00	0.0	0.0	0.0007	0.0351	0.0000
C	100.720	0.00	0.0	0.0	0.0013	0.0669	0.0000
C	95.000	0.00	0.0	0.0	0.0003	0.0172	0.0000
C	94.000	0.00	0.0	0.0	0.0144	0.8702	0.0000
C	85.000	0.00	0.0	0.0	0.0002	0.0172	0.0000
C	75.000	0.00	0.0	0.0	0.0002	0.0172	0.0000
C	71.500	0.00	0.0	0.0	0.1407	14.6419	0.0000
C	65.000	0.00	0.0	0.0	0.0001	0.0172	0.0000
C	55.000	0.00	0.0	0.0	0.0001	0.0172	0.0000
C	45.000	0.00	0.0	0.0	0.0001	0.0172	0.0000
C	35.000	0.00	0.0	0.0	0.0000	0.0172	0.0000
C	26.620	0.00	0.0	0.0	0.0222	16.6612	0.0000
C	25.000	0.00	0.0	0.0	0.0000	0.0172	0.0000
C	15.000	0.00	0.0	0.0	0.0000	0.0172	0.0000

D	179.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000
D	0.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000

ANTENNA LOADING
=====

.....ANTENNA.....	ATTACHMENT		ANTENNA FORCES.....					
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.00	0.00	0.00	0.00

LOADING CONDITION AL

Seismic - Azimuth: 0° (0.9 D - 1.0 Ev + 1.0 Eh)

LOADS ON POLE
=====

LOAD	ELEV	APPLY..LOAD..AT	LOAD	FORCES.....		MOMENTS.....	
TYPE	ft	RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL
		ft			kip	kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0751	0.9224	0.0000
C	175.000	0.00	0.0	0.0	0.1423	1.7481	0.0000
C	174.500	0.00	0.0	0.0	0.0009	0.0109	0.0000
C	165.000	0.00	0.0	0.0	0.0887	1.2256	0.0000
C	165.000	0.00	0.0	0.0	0.0186	0.2566	0.0000
C	165.000	0.00	0.0	0.0	0.0697	0.9640	0.0000
C	165.000	0.00	0.0	0.0	0.0009	0.0122	0.0000
C	158.620	0.00	0.0	0.0	0.3160	4.7266	0.0000
C	155.000	0.00	0.0	0.0	0.0008	0.0122	0.0000
C	154.000	0.00	0.0	0.0	0.4386	6.9584	0.0000
C	154.000	0.00	0.0	0.0	0.0527	0.8359	0.0000
C	145.000	0.00	0.0	0.0	0.0007	0.0122	0.0000
C	135.000	0.00	0.0	0.0	0.0006	0.0122	0.0000
C	125.000	0.00	0.0	0.0	0.0005	0.0122	0.0000
C	117.750	0.00	0.0	0.0	0.2705	7.3402	0.0000
C	115.720	0.00	0.0	0.0	0.0009	0.0248	0.0000
C	115.720	0.00	0.0	0.0	0.0019	0.0544	0.0000
C	115.000	0.00	0.0	0.0	0.0004	0.0122	0.0000
C	114.000	0.00	0.0	0.0	0.0475	1.3750	0.0000
C	114.000	0.00	0.0	0.0	0.0185	0.5354	0.0000
C	109.000	0.00	0.0	0.0	0.0014	0.0435	0.0000
C	109.000	0.00	0.0	0.0	0.0012	0.0379	0.0000
C	109.000	0.00	0.0	0.0	0.0194	0.6152	0.0000
C	105.000	0.00	0.0	0.0	0.0004	0.0122	0.0000
C	100.720	0.00	0.0	0.0	0.0007	0.0248	0.0000
C	100.720	0.00	0.0	0.0	0.0013	0.0474	0.0000
C	95.000	0.00	0.0	0.0	0.0003	0.0122	0.0000
C	94.000	0.00	0.0	0.0	0.0144	0.6152	0.0000
C	85.000	0.00	0.0	0.0	0.0002	0.0122	0.0000
C	75.000	0.00	0.0	0.0	0.0002	0.0122	0.0000
C	71.500	0.00	0.0	0.0	0.1407	10.3525	0.0000
C	65.000	0.00	0.0	0.0	0.0001	0.0122	0.0000
C	55.000	0.00	0.0	0.0	0.0001	0.0122	0.0000
C	45.000	0.00	0.0	0.0	0.0001	0.0122	0.0000
C	35.000	0.00	0.0	0.0	0.0000	0.0122	0.0000
C	26.620	0.00	0.0	0.0	0.0222	11.7801	0.0000
C	25.000	0.00	0.0	0.0	0.0000	0.0122	0.0000
C	15.000	0.00	0.0	0.0	0.0000	0.0122	0.0000
D	179.000	0.00	180.0	180.0	0.0000	0.0000	0.0000
D	0.000	0.00	180.0	180.0	0.0000	0.0000	0.0000

ANTENNA LOADING
=====

.....ANTENNA.....	ATTACHMENT		ANTENNA FORCES.....					
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.00	0.00	0.00	0.00

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180' Monopole / US-NY-2029, NY

MAXIMUM POLE DEFORMATIONS CALCULATED(w.r.t. wind direction)

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MAST ELEV ft	DEFLECTIONS (ft)			ROTATIONS (deg)		
	HORIZONTAL ALONG	ACROSS	DOWN	TILT ALONG	ACROSS	TWIST
179.0	9.67H	-0.19H	0.66F	5.06H	0.09U	-0.01H
174.1	9.23H	-0.18H	0.62F	5.06H	0.09U	-0.01H
169.1	8.80H	-0.17H	0.58F	5.05H	0.09U	-0.01H
164.2	8.37H	-0.17H	0.54F	5.04H	0.09U	-0.01H
159.3	7.94H	-0.16H	0.51F	5.02H	0.09U	-0.01H
154.4	7.52H	-0.15H	0.47F	4.98H	0.09U	-0.01H
149.4	7.09H	-0.14H	0.43F	4.93H	-0.09H	-0.01H
144.5	6.67H	-0.13H	0.40F	4.87H	-0.09H	-0.01H
138.2	6.15H	-0.12H	0.36F	4.76H	-0.09H	-0.01H
132.5	5.69H	-0.12H	0.32F	4.64H	-0.09H	-0.01H
126.8	5.23H	-0.11H	0.28F	4.50H	-0.09H	-0.01H
121.1	4.79H	-0.10H	0.25F	4.35H	-0.08H	-0.01H
115.4	4.37H	-0.09H	0.22F	4.18H	-0.08H	-0.01H
109.7	3.97H	-0.08H	0.19A	4.01H	-0.08H	-0.01H
104.0	3.58H	-0.07H	0.16A	3.82H	-0.08H	-0.01H
98.2	3.21H	-0.07H	0.14A	3.62H	-0.07H	-0.01H
91.0	2.76H	-0.06H	0.11A	3.40H	-0.07H	-0.01H
85.6	2.45H	-0.05H	0.09A	3.22H	-0.07H	-0.01H
80.2	2.16H	-0.05H	0.08A	3.03H	-0.06H	0.00H
74.8	1.89H	-0.04H	0.06A	2.85H	-0.06H	0.00H
69.4	1.63H	-0.03H	0.05A	2.65H	-0.05H	0.00H
64.0	1.39F	-0.03H	0.04A	2.46H	-0.05H	0.00H
58.6	1.17F	-0.02H	0.03A	2.26H	-0.05H	0.00H
53.2	0.96F	-0.02H	0.02A	2.06H	-0.04H	0.00H
44.7	0.68F	-0.01H	0.01A	1.74F	-0.04H	0.00H
39.2	0.52F	-0.01H	0.01A	1.53F	-0.03H	0.00H
33.6	0.38F	-0.01H	0.01A	1.31F	-0.03H	0.00H
28.0	0.27F	-0.01H	0.00A	1.09F	-0.02H	0.00H
22.4	0.17F	0.00H	0.00A	0.87F	-0.02H	0.00H
16.8	0.10F	0.00H	0.00A	0.65F	-0.01H	0.00H

11.2	0.04F	0.00H	0.00AE	0.43F	-0.01H	0.00H
5.6	0.01F	0.00H	0.00AD	0.22F	0.00H	0.00H
0.0	0.00A	0.00A	0.00A	0.00A	0.00A	0.00A

MAXIMUM ANTENNA AND REFLECTOR ROTATIONS

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ELEV	ANT	ANT	 BEAM DEFLECTIONS (deg)			
AZI	TYPE		ROLL	YAW	PITCH	TOTAL
ft	deg					
165.0	0.0	HP	4.964 D	0.192 C	-4.999 G	4.999 G

MAXIMUM POLE FORCES CALCULATED(w.r.t. to wind direction)

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MAST	TOTAL	SHEAR.w.r.t.WIND.DIR	MOMENT.w.r.t.WIND.DIR			TORSION
ELEV	AXIAL	ALONG	ALONG	ACROSS	ACROSS	
ft	kip	kip	ft-kip	ft-kip	ft-kip	ft-kip
179.0	-0.08 A	-0.21 G	-0.14 B	0.60 P	-0.36 Q	0.03 B
	9.41 AD	6.87 J	-0.14 B	-7.69 K	0.35 B	-0.04 B
174.1	9.41 AD	6.95 U	-0.24 O	-7.66 O	-0.39 O	-0.04 B
	10.58 AD	7.39 U	-0.24 O	-43.97 P	1.01 W	-0.11 B
169.1	10.59 AD	7.58 P	-0.35 O	-44.19 P	0.86 W	-0.13 B
	17.85 AD	14.04 X	0.45 B	-88.60 P	2.34 O	0.66 D
164.2	17.86 AD	14.10 P	-0.45 E	-88.94 P	2.99 O	0.66 D
	19.08 AD	14.55 P	-0.45 E	-161.92 P	3.35 E	0.77 Q
159.3	19.08 AD	14.59 G	-0.58 E	-161.97 P	3.19 E	0.78 Q
	20.35 AD	15.08 G	-0.58 E	-237.18 G	6.08 E	0.91 Q
154.4	20.35 AD	15.20 X	0.47 U	-237.48 G	6.28 E	0.94 Q
	46.38 AD	26.52 X	0.47 U	-370.83 G	7.38 E	1.07 Q
149.4	46.38 AD	26.46 X	0.55 C	-370.54 G	7.32 E	1.08 Q
	47.70 AD	26.96 X	0.55 C	-509.34 G	-8.01 U	1.15 Q
144.5	47.70 AD	27.21 H	0.72 U	-509.20 G	-7.99 U	1.14 Q
	50.46 AD	27.83 H	0.72 U	-689.09 G	-12.62 U	-1.30 H
138.2	50.46 AD	27.83 H	0.69 B	-688.98 G	-12.58 U	-1.29 H
	52.07 AD	28.43 H	0.69 B	-859.70 H	-16.48 U	-1.60 H
132.5	52.07 AD	28.31 F	0.67 B	-859.58 H	-16.65 U	-1.61 H
	53.68 AD	28.89 F	0.67 B	-1033.66 H	-20.28 U	-1.91 H
126.8	53.68 AD	29.00 H	0.92 U	-1033.23 H	-20.26 U	-1.92 H
	55.35 AD	29.61 H	0.92 U	-1211.83 H	-25.63 U	-2.25 H
121.1	55.35 AD	29.45 F	0.74 U	-1211.68 H	-25.70 U	-2.25 H
	57.18 AD	30.21 F	0.74 U	-1391.90 H	-30.05 U	-2.59 H
115.4	57.18 AD	30.12 H	0.76 B	-1391.89 H	-29.78 U	-2.60 H
	67.27 AD	35.44 H	0.76 B	-1597.81 H	-34.11 U	-2.93 H
109.7						

	67.27	AD	35.49	H	-0.73	H	-1597.87	H	-34.08	U	-2.95	H
	70.44	AD	36.77	H	-0.73	H	-1817.84	H	-36.99	U	-3.28	H
104.0	70.44	AD	36.88	H	0.87	O	-1817.94	H	-36.99	U	-3.27	H
	72.35	AD	37.66	H	0.87	O	-2042.52	H	-40.52	U	-3.62	H
98.2	72.35	AD	37.52	H	-0.84	H	-2042.59	H	-40.53	U	-3.62	H
	77.65	AD	38.82	H	-0.84	H	-2334.81	H	-45.04	U	-4.04	H
91.0	77.65	AD	38.91	H	0.86	O	-2334.07	H	-45.12	U	-4.05	H
	79.62	AD	39.49	H	0.86	O	-2557.58	H	-48.14	U	-4.35	H
85.6	79.62	AD	39.40	F	-1.03	H	-2557.37	H	-48.04	U	-4.35	H
	81.64	AD	40.01	F	-1.03	H	-2782.03	H	52.13	H	-4.69	H
80.2	81.65	AD	39.97	H	1.06	O	-2781.95	H	52.15	H	-4.69	H
	83.70	AD	40.58	H	1.06	O	-3010.12	H	57.61	H	-5.00	H
74.8	83.70	AD	40.62	A	-1.03	H	-3010.30	H	57.61	H	-5.00	H
	85.75	AD	41.20	A	-1.03	H	-3240.18	H	63.12	H	-5.29	H
69.4	85.75	AD	41.22	A	-1.06	H	-3240.21	H	63.04	H	-5.28	H
	87.86	AD	41.84	A	-1.06	H	-3473.61	H	68.70	H	-5.56	H
64.0	87.86	AD	41.77	A	-0.97	H	-3473.62	H	68.66	H	-5.56	H
	89.99	AD	42.36	A	-0.97	H	-3709.51	H	73.83	H	-5.80	H
58.6	89.99	AD	42.39	A	1.04	O	-3709.25	H	73.96	H	-5.80	H
	92.17	AD	43.01	A	1.04	O	-3947.83	H	78.74	H	-6.01	H
53.2	92.17	AD	43.03	A	-1.03	H	-3947.81	H	78.76	H	-6.01	H
	98.08	AD	43.98	A	-1.03	H	-4328.81	H	87.47	H	-6.33	H
44.7	98.08	AD	43.89	A	-1.10	H	-4328.72	H	87.47	H	-6.33	H
	100.39	AD	44.49	A	-1.10	H	-4582.86	H	93.58	H	-6.51	H
39.2	100.39	AD	44.56	A	-1.14	H	-4582.84	H	93.60	H	-6.51	H
	102.75	AD	45.17	A	-1.14	H	-4839.89	F	99.94	H	-6.68	H
33.6	102.75	AD	45.22	A	-1.11	H	-4839.81	F	99.96	H	-6.68	H
	105.10	AD	45.79	A	-1.11	H	-5099.91	F	106.13	H	-6.82	H
28.0	105.10	AD	45.77	A	-1.14	H	-5099.87	F	106.10	H	-6.82	H
	107.51	AD	46.35	A	-1.14	H	-5361.72	F	112.46	H	-6.94	H
22.4	107.51	AD	46.33	A	-1.15	H	-5361.73	F	112.46	H	-6.94	H
	109.90	AD	46.88	A	-1.15	H	-5625.61	F	118.90	H	-7.03	H
16.8	109.90	AD	46.96	A	-1.13	H	-5625.61	F	118.88	H	-7.03	H
	112.35	AD	47.52	A	-1.13	H	-5891.65	F	125.19	H	-7.09	H
11.2	112.35	AD	47.47	A	-1.12	H	-5891.64	F	125.19	H	-7.09	H
	114.78	AD	48.00	A	-1.12	H	-6159.46	F	131.45	H	-7.13	H
5.6	114.78	AD	48.03	A	-1.11	H	-6159.46	F	131.45	H	-7.13	H
	117.20	AD	48.56	A	-1.11	H	-6428.91	F	137.68	H	-7.14	H
base	117.20	AD	-48.56	A	1.11	H	6428.91	F	-137.68	H	7.14	H
reaction												

COMPLIANCE WITH 4.8.2 & 4.5.4

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ELEV ft	AXIAL	BENDING	SHEAR + TORSIONAL	TOTAL	SATISFIED	D/t (w/t)	MAX ALLOWED
179.00	0.00A	0.00P	0.00G	0.00P	YES	19.26A	45.2
174.07	0.00AD	0.00K	0.01J	0.01K	YES	19.79A	45.2
	0.00AD	0.00O	0.01U	0.01F	YES	19.79A	45.2
169.14	0.00AD	0.02P	0.01U	0.02E	YES	20.33A	45.2
	0.00AD	0.02P	0.01P	0.02J	YES	20.33A	45.2
164.21	0.01AD	0.04P	0.01X	0.05H	YES	20.86A	45.2
	0.01AD	0.04P	0.01P	0.05H	YES	20.86A	45.2
159.29	0.01AD	0.07P	0.01P	0.08F	YES	21.40A	45.2
	0.01AD	0.07P	0.01G	0.08F	YES	21.40A	45.2
154.36	0.01AD	0.10G	0.01G	0.11G	YES	21.93A	45.2
	0.01AD	0.10G	0.01X	0.11G	YES	21.93A	45.2
149.43	0.02AD	0.16G	0.02X	0.17G	YES	22.47A	45.2
	0.02AD	0.16G	0.02X	0.17G	YES	22.47A	45.2
144.50	0.02AD	0.21G	0.02X	0.22G	YES	23.00A	45.2
	0.02AD	0.21G	0.02H	0.22G	YES	23.00A	45.2
138.25	0.02AD	0.27G	0.02H	0.28G	YES	23.68A	45.2
	0.02AD	0.28G	0.02H	0.28G	YES	23.33A	45.2
132.54	0.02AD	0.33H	0.02H	0.34H	YES	23.95A	45.2
	0.02AD	0.33H	0.02F	0.34H	YES	23.95A	45.2
126.82	0.02AD	0.38H	0.02F	0.39H	YES	24.57A	45.2
	0.02AD	0.38H	0.02H	0.39H	YES	24.57A	45.2
121.11	0.02AD	0.43H	0.02H	0.44H	YES	25.19A	45.2
	0.02AD	0.43H	0.02F	0.44H	YES	25.19A	45.2
115.39	0.02AD	0.48H	0.02F	0.49H	YES	25.81A	45.2
	0.02AD	0.48H	0.02H	0.49H	YES	25.81A	45.2
109.68	0.02AD	0.53H	0.02H	0.54H	YES	26.43A	45.2
	0.02AD	0.53H	0.02H	0.54H	YES	26.43A	45.2
103.96	0.02AD	0.58H	0.02H	0.59H	YES	27.05A	45.2
	0.02AD	0.58H	0.02H	0.59H	YES	27.05A	45.2
98.25	0.02AD	0.63H	0.02H	0.64H	YES	27.67A	45.2
	0.02AD	0.49H	0.02H	0.50H	YES	23.00A	45.2
91.00	0.02AD	0.53H	0.02H	0.54H	YES	23.66A	45.2
	0.02AD	0.54H	0.02H	0.55H	YES	23.37A	45.2
85.61	0.02AD	0.58H	0.02H	0.59H	YES	23.85A	45.2
	0.02AD	0.58H	0.02F	0.59H	YES	23.85A	45.2
80.21	0.02AD	0.61H	0.02F	0.62H	YES	24.34A	45.2
	0.02AD	0.61H	0.02H	0.62H	YES	24.34A	45.2

74.82	0.02AD	0.64H	0.02H	0.65H	YES	24.83A	45.2
	0.02AD	0.64H	0.02F	0.65H	YES	24.83A	45.2
69.43	0.02AD	0.67H	0.02F	0.68H	YES	25.32A	45.2
	0.02AD	0.67H	0.02H	0.68H	YES	25.32A	45.2
64.04	0.02AD	0.69H	0.02H	0.71H	YES	25.81A	45.2
	0.02AD	0.69H	0.02F	0.71H	YES	25.81A	45.2
58.64	0.02AD	0.72H	0.02H	0.73H	YES	26.29A	45.2
	0.02AD	0.72H	0.02H	0.73H	YES	26.29A	45.2
53.25	0.02AD	0.75H	0.02H	0.76H	YES	26.78A	45.2
	0.02AD	0.75H	0.02H	0.76H	YES	26.78A	45.2
44.75	0.02AD	0.78H	0.02H	0.80H	YES	27.55A	45.2
	0.02AD	0.80H	0.02F	0.81H	YES	27.20A	45.2
39.16	0.02AD	0.82H	0.02H	0.84H	YES	27.71A	45.2
	0.02AD	0.82H	0.02H	0.84H	YES	27.71A	45.2
33.56	0.02AD	0.85H	0.02H	0.86H	YES	28.21A	45.2
	0.02AD	0.85H	0.02H	0.86H	YES	28.21A	45.2
27.97	0.02AD	0.87H	0.02H	0.88H	YES	28.72A	45.2
	0.02AD	0.87H	0.02H	0.88H	YES	28.72A	45.2
22.37	0.02AD	0.89H	0.02H	0.90H	YES	29.22A	45.2
	0.02AD	0.89H	0.02H	0.90H	YES	29.22A	45.2
16.78	0.02AD	0.91H	0.02H	0.93H	YES	29.73A	45.2
	0.02AD	0.91H	0.02H	0.93H	YES	29.73A	45.2
11.19	0.02AD	0.93H	0.02H	0.95H	YES	30.24A	45.2
	0.02AD	0.93H	0.02H	0.95H	YES	30.24A	45.2
5.59	0.02AD	0.95H	0.02H	0.97H	YES	30.74A	45.2
	0.02AD	0.95H	0.02H	0.97H	YES	30.74A	45.2
0.00	0.02AD	0.97H	0.02H	0.99H	YES	31.25A	45.2

MAXIMUM LOADS ONTO FOUNDATION (w.r.t. wind direction)

DOWN	SHEAR.w.r.t.WIND.DIR		MOMENT.w.r.t.WIND.DIR		TORSION
kip	ALONG	ACROSS	ALONG	ACROSS	ft-kip
	kip	kip	ft-kip	ft-kip	
117.20	48.56	-1.11	-6428.91	137.68	-7.14
AD	A	H	F	H	H

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180' Monopole / US-NY-2029, NY

***** Service Load Condition *****

* Only 1 condition(s) shown in full

* RRUs/TMAs were assumed to be behind antennas

* Some concentrated wind loads may have been derived from full-scale wind tunnel testing

LOADING CONDITION A =====

60 mph wind with no ice. Wind Azimuth: 0° (1.0 D + 1.0 Wo)

LOADS ON POLE

=====

LOAD TYPE	ELEV ft	APPLY..LOAD..AT RADIUS ft	..AZI AZI	LOAD AZI	FORCES..... HORIZ kip	DOWN kip	MOMENTS..... VERTICAL ft-kip	TORSNAL ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.0605	0.0000	0.0000
C	175.000	0.00	0.0	0.0	1.6928	2.0098	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0079	0.0126	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.4091	0.0000	0.0000
C	165.000	0.00	0.0	0.0	1.5235	1.1083	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0087	0.0140	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0086	0.0140	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	0.9610	0.0000	0.0000
C	154.000	0.00	0.0	0.0	2.9433	8.0000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0085	0.0140	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0083	0.0140	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0082	0.0140	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0625	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0469	0.0285	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0081	0.0140	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.6156	0.0000	0.0000
C	114.000	0.00	0.0	0.0	1.2732	1.5808	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0436	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0378	0.0500	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1366	0.7073	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0079	0.0140	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0544	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0456	0.0285	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0078	0.0140	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.1324	0.7073	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0076	0.0140	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0074	0.0140	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0072	0.0140	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0069	0.0140	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0066	0.0140	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0063	0.0140	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0059	0.0140	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0053	0.0140	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0236	0.1222	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0262	0.1412	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0267	0.2877	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0267	0.2877	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0268	0.1468	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0290	0.1689	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0293	0.3782	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0293	0.3782	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0292	0.2100	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.0298	0.2309	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.0297	0.2351	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0297	0.2351	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0296	0.4781	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0296	0.4781	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0293	0.2431	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0256	0.2648	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0257	0.2691	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0261	0.2735	0.0000	0.0000

ANTENNA LOADING

=====

.....ANTENNA.....		ATTACHMENT		ANTENNA FORCES.....				
TYPE	ELEV ft	AZI	RAD ft	AZI	AXIAL kip	SHEAR kip	GRAVITY kip	TORSION ft-kip
HP	165.0	0.0	2.3	0.0	0.13	0.00	0.11	0.00

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MAXIMUM POLE DEFORMATIONS CALCULATED(w.r.t. wind direction)

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MAST ELEV ft	DEFLECTIONS (ft).....			ROTATIONS (deg).....		
	 HORIZONTAL ALONG	 ACROSS	DOWN	 TILT ALONG	 ACROSS	TWIST
179.0	2.61C	-0.04F	0.05B	1.36C	-0.02F	0.00J
174.1	2.50C	-0.04F	0.05B	1.36C	-0.02F	0.00J
169.1	2.38C	-0.04F	0.05B	1.36C	-0.02F	0.00J
164.2	2.26C	-0.04F	0.04B	1.35C	-0.02F	0.00J
159.3	2.15C	-0.03F	0.04B	1.35C	-0.02F	0.00J
154.4	2.03C	-0.03F	0.04B	1.34C	-0.02F	0.00J
149.4	1.92C	-0.03F	0.03B	1.33C	-0.02F	0.00J
144.5	1.80C	-0.03F	0.03B	1.31C	-0.02F	0.00J
138.2	1.66C	-0.03F	0.03B	1.28C	-0.02F	0.00J
132.5	1.54C	-0.03F	0.03B	1.25C	-0.02F	0.00J
126.8	1.42C	-0.02F	0.02C	1.21C	-0.02F	0.00J
121.1	1.30C	-0.02F	0.02C	1.17C	-0.02F	0.00J
115.4	1.18C	-0.02F	0.02C	1.13C	-0.02F	0.00J
109.7	1.07C	-0.02F	0.02C	1.08C	-0.02F	0.00J
104.0	0.97C	-0.02F	0.01C	1.03C	-0.02F	0.00J
98.2	0.87C	-0.01F	0.01C	0.98C	-0.02F	0.00J
91.0	0.75C	-0.01F	0.01C	0.92C	-0.02F	0.00J
85.6	0.66C	-0.01F	0.01C	0.87C	-0.01F	0.00J
80.2	0.58C	-0.01F	0.01C	0.82C	-0.01F	0.00J
74.8	0.51C	-0.01F	0.01C	0.77C	-0.01F	0.00J
69.4	0.44C	-0.01F	0.01C	0.72C	-0.01F	0.00J
64.0	0.37C	-0.01F	0.00C	0.66C	-0.01F	0.00J
58.6	0.31C	-0.01F	0.00C	0.61C	-0.01F	0.00J
53.2	0.26C	0.00F	0.00C	0.56C	-0.01F	0.00J
44.7	0.18C	0.00F	0.00C	0.47C	-0.01F	0.00J
39.2	0.14C	0.00F	0.00C	0.41C	-0.01F	0.00J
33.6	0.10C	0.00F	0.00C	0.35C	-0.01F	0.00J
28.0	0.07C	0.00F	0.00C	0.29C	0.00F	0.00J
22.4	0.05C	0.00F	0.00C	0.24C	0.00F	0.00J
16.8	0.03C	0.00F	0.00C	0.18C	0.00F	0.00J
11.2	0.01C	0.00F	0.00C	0.12C	0.00F	0.00J
5.6	0.00C	0.00F	0.00F	0.06C	0.00F	0.00J

0.0 0.00A 0.00A 0.00A 0.00A 0.00A 0.00A

MAXIMUM ANTENNA AND REFLECTOR ROTATIONS

=====

ELEV	ANT	ANT	 BEAM DEFLECTIONS (deg)			
ft	AZI	TYPE	ROLL	YAW	PITCH	TOTAL
	deg					
165.0	0.0	HP	-1.344 J	0.014 K	1.331 A	1.331 A

MAXIMUM POLE FORCES CALCULATED(w.r.t. to wind direction)

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MAST	TOTAL	SHEAR.w.r.t.WIND.DIR	MOMENT.w.r.t.WIND.DIR	TORSION	
ELEV	AXIAL	ALONG	ALONG	ACROSS	ACROSS
ft	kip	kip	ft-kip	ft-kip	ft-kip
179.0	0.01 F	0.03 I	-0.04 F	-0.08 A	0.08 C
	3.70 F	1.85 I	-0.04 F	-2.16 F	0.13 F
174.1	3.70 L	1.86 K	0.05 H	-2.15 F	0.11 F
	4.32 L	1.98 K	0.05 H	-11.90 F	-0.22 H
169.1	4.33 K	2.01 B	-0.06 F	-12.01 L	0.23 L
	7.61 K	3.78 B	0.09 C	-23.87 E	0.45 F
164.2	7.61 K	3.81 B	-0.15 K	-23.84 E	0.46 F
	8.26 K	3.93 B	-0.15 K	-43.42 E	0.69 F
159.3	8.26 F	3.94 B	-0.12 K	-43.49 E	0.68 E
	8.93 F	4.07 B	-0.12 K	-63.86 B	0.94 K
154.4	8.94 F	4.08 B	-0.19 K	-63.94 E	0.96 K
	18.57 F	7.15 B	-0.19 K	-99.58 B	1.90 K
149.4	18.58 F	7.13 B	-0.16 K	-99.50 B	1.87 K
	19.28 F	7.27 B	-0.16 K	-137.10 B	2.67 K
144.5	19.28 F	7.26 B	-0.19 B	-137.06 B	2.68 K
	21.08 F	7.43 B	-0.19 B	-185.74 B	3.74 K
138.2	21.08 F	7.44 B	-0.16 B	-185.67 B	3.76 K
	21.94 F	7.60 B	-0.16 B	-231.35 B	4.42 K
132.5	21.94 F	7.57 C	-0.16 B	-231.44 B	4.43 K
	22.81 F	7.72 C	-0.16 B	-277.63 B	5.21 K
126.8	22.81 F	7.75 C	-0.19 B	-277.67 B	5.27 K
	23.71 F	7.92 C	-0.19 B	-324.80 B	5.83 K
121.1	23.71 F	7.95 C	-0.19 B	-324.70 B	5.86 K
	24.70 F	8.16 C	-0.19 B	-372.58 B	6.42 K
115.4	24.70 F	8.16 C	-0.20 F	-372.45 B	6.43 K
	27.83 F	9.61 C	-0.20 F	-427.50 B	6.99 K
109.7	27.83 F	9.61 C	-0.18 F	-427.50 B	6.99 K
	29.58 F	9.95 C	-0.18 F	-486.02 B	7.66 K
104.0					

	29.58 F	9.94 C	-0.20 F	-485.98 B	7.64 K	-0.18 K
	30.62 F	10.16 C	-0.20 F	-545.94 B	8.59 F	-0.19 K
98.2	30.62 F	10.21 C	-0.18 F	-545.95 B	8.63 F	-0.19 K
	34.08 F	10.56 C	-0.18 F	-624.58 C	10.02 F	-0.19 K
91.0	34.08 F	10.57 C	-0.20 F	-624.60 C	9.97 F	-0.19 K
	35.22 F	10.73 C	-0.20 F	-684.89 C	11.07 F	-0.19 K
85.6	35.22 F	10.68 C	-0.21 F	-684.92 C	11.07 F	-0.19 K
	36.40 F	10.84 C	-0.21 F	-745.75 C	12.25 F	-0.19 K
80.2	36.40 F	10.87 C	-0.20 F	-745.78 C	12.21 F	-0.19 K
	37.59 F	11.04 C	-0.20 F	-807.59 C	13.33 F	-0.19 K
74.8	37.59 F	11.04 C	-0.21 F	-807.63 C	13.32 F	-0.19 K
	38.79 F	11.20 C	-0.21 F	-870.26 C	14.48 F	-0.19 K
69.4	38.79 F	11.21 C	-0.21 F	-870.31 C	14.45 F	-0.19 K
	40.02 F	11.37 C	-0.21 F	-933.65 C	15.59 F	-0.18 K
64.0	40.02 F	11.34 F	-0.19 F	-933.69 C	15.60 F	-0.18 K
	41.25 F	11.50 F	-0.19 F	-997.68 C	16.68 F	-0.18 J
58.6	41.25 F	11.51 C	-0.19 F	-997.71 C	16.70 F	-0.18 J
	42.54 F	11.68 C	-0.19 F	-1062.48 C	17.76 F	-0.18 J
53.2	42.54 F	11.70 F	-0.20 F	-1062.51 C	17.76 F	-0.18 J
	46.61 F	11.96 F	-0.20 F	-1166.24 C	19.51 F	-0.18 J
44.7	46.61 F	11.96 F	-0.20 F	-1166.23 C	19.50 F	-0.18 J
	47.98 F	12.12 F	-0.20 F	-1235.50 C	20.62 F	-0.18 J
39.2	47.98 F	12.12 F	-0.19 F	-1235.51 C	20.61 F	-0.18 J
	49.39 F	12.29 F	-0.19 F	-1305.49 C	21.71 F	-0.18 J
33.6	49.39 F	12.31 C	-0.19 F	-1305.52 C	21.72 F	-0.18 J
	50.80 F	12.47 C	-0.19 F	-1376.40 C	22.78 F	-0.18 J
28.0	50.80 F	12.47 C	-0.19 F	-1376.41 C	22.79 F	-0.18 J
	52.24 F	12.63 C	-0.19 F	-1447.93 C	23.87 F	-0.18 J
22.4	52.24 F	12.64 C	-0.19 F	-1447.93 C	23.88 F	-0.18 J
	53.69 F	12.79 C	-0.19 F	-1520.08 C	24.93 F	-0.18 J
16.8	53.69 F	12.79 C	-0.20 F	-1520.08 C	24.93 F	-0.18 J
	55.18 F	12.94 C	-0.20 F	-1592.82 C	26.04 F	-0.18 J
11.2	55.18 F	12.94 C	-0.20 F	-1592.81 C	26.04 F	-0.18 J
	56.69 F	13.08 C	-0.20 F	-1666.06 C	27.16 F	-0.18 J
5.6	56.69 F	13.08 C	-0.20 F	-1666.06 C	27.16 F	-0.18 J
	58.21 F	13.22 C	-0.20 F	-1739.79 C	28.30 F	-0.18 J

base						
reaction	58.21 F	-13.22 C	0.20 F	1739.79 C	-28.30 F	0.18 J

COMPLIANCE WITH 4.8.2 & 4.5.4
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ELEV	AXIAL	BENDING	SHEAR +	TOTAL SATISFIED	D/t (w/t)	MAX
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ft	TORSIONAL					ALLOWED	
179.00	0.00F	0.00I	0.00C	0.00I	YES	19.26A	45.2
	0.00F	0.00F	0.00I	0.00F	YES	19.79A	45.2
174.07	0.00L	0.00F	0.00K	0.00F	YES	19.79A	45.2
	0.00L	0.01F	0.00K	0.01F	YES	20.33A	45.2
169.14	0.00K	0.01L	0.00B	0.01L	YES	20.33A	45.2
	0.00K	0.01E	0.00B	0.01E	YES	20.86A	45.2
164.21	0.00K	0.01E	0.00B	0.01E	YES	20.86A	45.2
	0.00K	0.02E	0.00B	0.02E	YES	21.40A	45.2
159.29	0.00F	0.02E	0.00B	0.02E	YES	21.40A	45.2
	0.00F	0.03B	0.00B	0.03B	YES	21.93A	45.2
154.36	0.00F	0.03E	0.00B	0.03E	YES	21.93A	45.2
	0.01F	0.04B	0.01B	0.05B	YES	22.47A	45.2
149.43	0.01F	0.04B	0.01B	0.05B	YES	22.47A	45.2
	0.01F	0.06B	0.01B	0.06B	YES	23.00A	45.2
144.50	0.01F	0.06B	0.01B	0.06B	YES	23.00A	45.2
	0.01F	0.07B	0.01B	0.08B	YES	23.68A	45.2
138.25	0.01F	0.07B	0.01B	0.08B	YES	23.33A	45.2
	0.01F	0.09B	0.01B	0.10B	YES	23.95A	45.2
132.54	0.01F	0.09B	0.01C	0.10B	YES	23.95A	45.2
	0.01F	0.10B	0.01C	0.11B	YES	24.57A	45.2
126.82	0.01F	0.10B	0.01C	0.11B	YES	24.57A	45.2
	0.01F	0.12B	0.01C	0.12B	YES	25.19A	45.2
121.11	0.01F	0.12B	0.01C	0.12B	YES	25.19A	45.2
	0.01F	0.13B	0.01C	0.14B	YES	25.81A	45.2
115.39	0.01F	0.13B	0.01C	0.14B	YES	25.81A	45.2
	0.01F	0.14B	0.01C	0.15B	YES	26.43A	45.2
109.68	0.01F	0.14B	0.01C	0.15B	YES	26.43A	45.2
	0.01F	0.15B	0.01C	0.16B	YES	27.05A	45.2
103.96	0.01F	0.15B	0.01C	0.16B	YES	27.05A	45.2
	0.01F	0.17B	0.01C	0.18B	YES	27.67A	45.2
98.25	0.01F	0.13B	0.01C	0.14B	YES	23.00A	45.2
	0.01F	0.14C	0.01C	0.15C	YES	23.66A	45.2
91.00	0.01F	0.15C	0.01C	0.15C	YES	23.37A	45.2
	0.01F	0.15C	0.01C	0.16C	YES	23.85A	45.2
85.61	0.01F	0.15C	0.01C	0.16C	YES	23.85A	45.2
	0.01F	0.16C	0.01C	0.17C	YES	24.34A	45.2
80.21	0.01F	0.16C	0.01C	0.17C	YES	24.34A	45.2
	0.01F	0.17C	0.01C	0.18C	YES	24.83A	45.2
74.82	0.01F	0.17C	0.01C	0.18C	YES	24.83A	45.2

69.43	0.01F	0.18C	0.01C	0.19C	YES	25.32A	45.2
	0.01F	0.18C	0.01C	0.19C	YES	25.32A	45.2
64.04	0.01F	0.19C	0.01C	0.20C	YES	25.81A	45.2
	0.01F	0.19C	0.01F	0.20C	YES	25.81A	45.2
58.64	0.01F	0.19C	0.01F	0.20C	YES	26.29A	45.2
	0.01F	0.19C	0.01C	0.20C	YES	26.29A	45.2
53.25	0.01F	0.20C	0.01C	0.21C	YES	26.78A	45.2
	0.01F	0.20C	0.01F	0.21C	YES	26.78A	45.2
44.75	0.01F	0.21C	0.01F	0.22C	YES	27.55A	45.2
	0.01F	0.22C	0.01F	0.23C	YES	27.20A	45.2
39.16	0.01F	0.22C	0.01F	0.23C	YES	27.71A	45.2
	0.01F	0.22C	0.01F	0.23C	YES	27.71A	45.2
33.56	0.01F	0.23C	0.01F	0.24C	YES	28.21A	45.2
	0.01F	0.23C	0.01C	0.24C	YES	28.21A	45.2
27.97	0.01F	0.23C	0.01C	0.25C	YES	28.72A	45.2
	0.01F	0.23C	0.01C	0.25C	YES	28.72A	45.2
22.37	0.01F	0.24C	0.01C	0.25C	YES	29.22A	45.2
	0.01F	0.24C	0.01C	0.25C	YES	29.22A	45.2
16.78	0.01F	0.25C	0.01C	0.26C	YES	29.73A	45.2
	0.01F	0.25C	0.01C	0.26C	YES	29.73A	45.2
11.19	0.01F	0.25C	0.01C	0.26C	YES	30.24A	45.2
	0.01F	0.25C	0.01C	0.26C	YES	30.24A	45.2
5.59	0.01F	0.26C	0.01C	0.27C	YES	30.74A	45.2
	0.01F	0.26C	0.01C	0.27C	YES	30.74A	45.2
0.00	0.01F	0.26C	0.01C	0.28C	YES	31.25A	45.2

MAXIMUM LOADS ONTO FOUNDATION(w.r.t. wind direction)

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DOWN	SHEAR.w.r.t.WIND.DIR		MOMENT.w.r.t.WIND.DIR		TORSION
	ALONG	ACROSS	ALONG	ACROSS	
kip	kip	kip	ft-kip	ft-kip	ft-kip
58.21	13.22	-0.20	-1739.79	28.30	-0.18
F	C	F	C	F	J

=====

Seismic Load Effects
Equivalent Lateral Force Procedure
ANSI/TIA-222-H

Parameters	Description	h _i (ft.)	w _i (kips)	W _n (kips)	Vertical Distribution of Seismic Forces			
					$\frac{w_i h_i^k}{\sum w_i h_i^k}$	$\frac{F_x \text{ or } E_h}{\sum F_x \text{ or } E_h}$ (kips)	$\frac{E_v}{\sum E_v}$ (kips)	$\frac{1.2 D + 1.0 E_v}{\sum (1.2 D + 1.0 E_v)}$ (kips)
Risk Category	II	175.00	1.0605	0.0000	32,477.8125	0.0751	0.0320	0.9224
	1.500	175.00	2.0098	2.0098	61,550.1250	0.1423	0.0607	1.7481
R	Step Bolts/Safety Climb Load	174.50	0.0126	0.0000	383.6732	0.0009	0.0004	0.0109
S _s	Line Deadload	165.00	1.4091	0.0000	38,362.7475	0.0887	0.0426	1.2256
S ₁	Mount/Antenna Load	165.00	0.2950	0.2950	8,031.3750	0.0186	0.0089	0.2566
Site Class	Mount/Antenna Load	165.00	1.1083	1.1083	30,173.4675	0.0697	0.0335	0.9640
T _L (sec)	Step Bolts/Safety Climb Load	165.00	0.0140	0.0000	381.1500	0.0009	0.0004	0.0122
F _a	Structure - Section 1	158.62	5.4341	0.0000	136,723.6101	0.3160	0.1641	4.7266
F _v	Step Bolts/Safety Climb Load	155.00	0.0140	0.0000	336.3500	0.0008	0.0004	0.0122
S _{MS}	Antenna Load	154.00	8.0000	8.0000	189,728.0000	0.4386	0.2416	6.9584
S _{M1}	Line Deadload	154.00	0.9610	0.0000	22,791.0760	0.0527	0.0290	0.8359
S _{PS}	Step Bolts/Safety Climb Load	145.00	0.0140	0.0000	294.3500	0.0007	0.0004	0.0122
S _{D1}	Step Bolts/Safety Climb Load	135.00	0.0140	0.0000	255.1500	0.0006	0.0004	0.0122
T _s	Step Bolts/Safety Climb Load	125.00	0.0140	0.0000	218.7500	0.0005	0.0004	0.0122
I _e	Structure - Section 2	117.75	8.4390	0.0000	117,007.2624	0.2705	0.2549	7.3402
Ω	Antenna Load	115.72	0.0285	0.0000	381.6469	0.0009	0.0009	0.0248
C _s	Line Deadload	115.72	0.0625	0.0000	836.9449	0.0019	0.0019	0.0544
E (ksi)	Step Bolts/Safety Climb Load	115.00	0.0140	0.0000	185.1500	0.0004	0.0004	0.0122
I _{top} (in ⁴)	Line Deadload	114.00	0.6156	0.0000	8,000.3376	0.0185	0.0186	0.5354
I _{bot} (in ⁴)	Mount/Antenna Load	114.00	1.5808	0.0000	20,544.0768	0.0475	0.0477	1.3750
I _{avg} (in ⁴)	Antenna Load	109.00	0.0500	0.0000	594.0500	0.0014	0.0015	0.0435
g (in/s ²)	Line Deadload	109.00	0.0436	0.0000	518.0116	0.0012	0.0013	0.0379
W _t (kips)	Mount Load	109.00	0.7073	0.0000	8,403.4313	0.0194	0.0214	0.6152
W _u (kips)	Step Bolts/Safety Climb Load	105.00	0.0140	0.0000	154.3500	0.0004	0.0004	0.0122
W _L (kips)	Antenna Load	100.72	0.0285	0.0000	289.1188	0.0007	0.0009	0.0248
L _p (in)	Line Deadload	100.72	0.0544	0.0000	551.8618	0.0013	0.0016	0.0474
f ₁ (Hertz)	Step Bolts/Safety Climb Load	95.00	0.0140	0.0000	126.3500	0.0003	0.0004	0.0122
T (sec)	Mount Load	94.00	0.7073	0.0000	6,249.7028	0.0144	0.0214	0.6152
k _e	Step Bolts/Safety Climb Load	85.00	0.0140	0.0000	101.1500	0.0002	0.0004	0.0122
V _s (kips)	Step Bolts/Safety Climb Load	75.00	0.0140	0.0000	78.7500	0.0002	0.0004	0.0122
Seismic Design Category	B	71.50	11.9021	0.0000	60,846.5107	0.1407	0.3594	10.3525
		65.00	0.0140	0.0000	59.1500	0.0001	0.0004	0.0122
	Step Bolts/Safety Climb Load	55.00	0.0140	0.0000	42.3500	0.0001	0.0004	0.0122
	Step Bolts/Safety Climb Load	45.00	0.0140	0.0000	28.3500	0.0001	0.0004	0.0122
	Step Bolts/Safety Climb Load	35.00	0.0140	0.0000	17.1500	0.0000	0.0004	0.0122
	Structure - Section 4	26.62	13.5435	0.0000	9,597.2546	0.0222	0.4090	11.7801
	Step Bolts/Safety Climb Load	25.00	0.0140	0.0000	8.7500	0.0000	0.0004	0.0122

Seismic Load Effects
Equivalent Lateral Force Procedure
ANSI/TIA-222-H

Description	h_i (ft.)	w_i (kips)	W_{ei} (kips)	Vertical Distribution of Seismic Forces			
				$\frac{w_i h_i^{ke}}{W_{ei}}$	$\frac{F_{sx} \text{ or } E_{hx}}{E_v}$ (kips)	$\frac{1.2 D + 1.0 E_v}{E_v}$ (kips)	$\frac{0.9 D - 1.0 E_v}{E_v}$ (kips)
Step Bolts/Safety Climb Load	15.00	0.0140	0.0000	3.1500	0.0000	0.0004	0.0122
Σ		58.28	11.4131	756.332.50	1.75	1.76	50.69

Round Base Plate and Anchor Rods, per ANSI/TIA 222-H

Pole Data

Diameter:	68.460	in (flat to flat)
Thickness:	0.375	in
Yield (Fy):	65	ksi
# of Sides:	18	"0" IF Round
Strength (Fu):	80	ksi

Reactions

Moment, Mu:	6428.91	ft-kips
Axial, Pu:	70.19	kips
Shear, Vu:	48.56	kips

Anchor Rod Data

Quantity:	18	
Diameter:	2.25	in
Rod Material:	A615	
Strength (Fu):	100	ksi
Yield (Fy):	75	ksi
BC Diam. (in):	75.5	BC Override:

Plate Data

Diameter (in):	81.25	Dia. Override:
Thickness:	2	in
Yield (Fy):	50	ksi
Eff Width/Rod:	12.07	in
Drain Hole:	2.625	in. diameter
Drain Location:	32.25	in. center of pole to center of drain hole
Center Hole:	56.5	in. diameter

Anchor Rod Results

(per 4.9.9)

Maximum Put:	224.15 Kips
Φt^*Rnt :	243.75 Kips
Vu:	2.70 Kips
Φv^*Rnv :	149.10 Kips
Tension Interaction Ratio:	0.85
Maximum Puc:	230.97 Kips
Φc^*Rnc :	268.39 Kips
Vu:	2.70 Kips
Φc^*Rnvc :	120.77 Kips
Compression Interaction Ratio:	0.86
Maximum Interaction Ratio:	86.1% Pass

Base Plate Results

Base Plate (Mu/Z):	42.5 ksi
Allowable Φ^*Fy :	45.0 ksi (per AISC)
Base Plate Interaction Ratio:	94.5% Pass

MAT FOUNDATION DESIGN BY SABRE INDUSTRIES

180' Monopole PTI SERVICES US-NY-2029, NY (25-3277-JDS-R1) 04/21/25 TTW

Overall Loads:

Factored Moment (ft-kips)	6428.91
Factored Axial (kips)	70.19
Factored Shear (kips)	48.56
Bearing Design Strength (ksf)	2.985
Water Table Below Grade (ft)	27.2
Width of Mat (ft)	29
Thickness of Mat (ft)	1.75
Depth to Bottom of Slab (ft)	6
Quantity of Bolts in Bolt Circle	18
Bolt Circle Diameter (in)	75.5
Effective Anchor	
Bolt Embedment (in)	66.5
Diameter of Pier (ft)	8
Ht. of Pier Above Ground (ft)	0.5
Ht. of Pier Below Ground (ft)	4.25
Quantity of Bars in Mat	51
Bar Diameter in Mat (in)	1.27
Area of Bars in Mat (in ²)	64.61
Spacing of Bars in Mat (in)	6.81
Quantity of Bars Pier	50
Bar Diameter in Pier (in)	1
Tie Bar Diameter in Pier (in)	0.625
Spacing of Ties (in)	4
Area of Bars in Pier (in ²)	39.27
Spacing of Bars in Pier (in)	5.51
f'c (ksi)	4.5
fy (ksi)	60
Unit Wt. of Soil (kcf)	0.117
Unit Wt. of Concrete (kcf)	0.15

Max. Net Bearing Press. (ksf) 2.97

Allowable Bearing Pressure (ksf)	1.99
Safety Factor	2.00
Ultimate Bearing Pressure (ksf)	3.98
Bearing Φ s	0.75

Minimum Pier Diameter (ft)	8.00
Equivalent Square b (ft)	7.09
Square Pier? (Y/N)	N

Recommended Spacing (in) 5 to 12

Minimum Pier A_s (in ²)	36.19
Recommended Spacing (in)	5 to 12

Volume of Concrete (yd³) 63.35**Two-Way Shear Action:**

Average d (in)	16.73
ϕv_c (ksi)	0.183
$\phi v_c = \phi(2 + 4/\beta_c)f_c^{1/2}$	0.302
$\phi v_c = \phi(\alpha_s d/b_o + 2)f_c^{1/2}$	0.183
$\phi v_c = \phi 4f_c^{1/2}$	0.201
Shear perimeter, b_o (in)	407.23
β_c	1

 v_u (ksi) 0.148

J (in ³)	1.185E+07
c + d (in)	101.81
0.40M _{sc} (ft-kips)	2663.8

One-Way Shear: ϕV_c (kips) 585.8 V_u (kips) 412.8**Stability:**

Overturning Design Strength (ft-k) 9242.8

Total Applied M (ft-k) 6744.6

Pier-Slab Transfer by Flexure:

b_{slab} (ft)	13.25		
ϕM_n (ft-kips)	4057.7	$0.60M_{sc}$ (ft-kips)	3995.7

Pier Design:

ϕV_n (kips)	1275.6	V_u (kips)	48.6
$\phi V_c = \phi 2(1 + N_u / (2000 A_g)) f'_c{}^{1/2} b_w d$	745.5		
V_s (kips)	706.9	*** $V_s \max = 4 f'_c{}^{1/2} b_w d$ (kips)	1978.3
Maximum Spacing (in)	7.62	(Only if Shear Ties are Required)	
Actual Hook Development (in)	15.46	Req'd Hook Development l_{dh} (in) - Tension	12.52
		Req'd Hook Development l_{dc} (in) - Compression	13.50

Flexure in Slab:

ϕM_n (ft-kips)	4440.5	M_u (ft-kips)	2996.2
a (in)	2.91		
Steel Ratio	0.01110		
β_1	0.825		
Maximum Steel Ratio (ρ_t)	0.0197		
Minimum Steel Ratio	0.0018		
Rebar Development in Pad (in)	123.00	Required Development in Pad (in)	34.08

Condition	1 is OK, 0 Fails
Maximum Soil Bearing Pressure	1
Pier Area of Steel	1
Pier Shear	1
Interaction Diagram	1
Two-Way Shear Action	1
One-Way Shear Action	1
Overtaking	1
Flexure	1
Steel Ratio	1
Length of Development in Pad	1
Hook Development	1
Anchor Bolt Pullout	1
Anchor Bolt Punching Shear	1

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LFile for Windows, Version 2019-11.009

Analysis of Individual Files and Drilled Shafts
Subjected to Lateral Loading Using the p-y Method
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Files Used for Analysis

Path to file locations:
\Program Files (x86)\Ensoft\Lpile2019\files\

Name of input data file:
25-3277-JDS-R1.lpl1d

Name of output report file:
25-3277-JDS-R1.lpl1o

Name of plot output file:
25-3277-JDS-R1.lpl1p

Name of runtime message file:
25-3277-JDS-R1.lpl1r

Date and Time of Analysis

Date: April 17, 2025

Time: 9:16:14

Problem Title

Site : US-NY-2029, NY

Tower : 180' Monopole

Prepared for : PTI SERVICES

Job Number : 25-3277-JDS-R1

Engineer : TTW

Program Options and Settings

Computational Options:

- Conventional Analysis
- Engineering Units Used for Data Input and Computations:
- US Customary System Units (pounds, feet, inches)

Analysis Control Options:

- Maximum number of iterations allowed = 999
- Deflection tolerance for convergence = 1.0000E-05 in
- Maximum allowable deflection = 100.0000 in
- Number of pile increments = 100

Loading Type and Number of Cycles of Loading:

- Static loading specified
- Use of p-y modification factors for p-y curves not selected
- Analysis uses layering correction (Method of Georgiadis)
- No distributed lateral loads are entered
- Loading by lateral soil movements acting on pile not selected
- Input of shear resistance at the pile tip not selected
- Input of moment resistance at the pile tip not selected
- Input of side resistance moment along pile not selected
- Computation of pile-head foundation stiffness matrix not selected
- Push-over analysis of pile not selected
- Buckling analysis of pile not selected

Output Options:

- Output files use decimal points to denote decimal symbols.
- Report only summary tables of pile-head deflection, maximum bending moment, and maximum shear force in output report file.
- No p-y curves to be computed and reported for user-specified depths
- Print using wide report formats

Pile Structural Properties and Geometry

- Number of pile sections defined = 1
- Total length of pile = 27.500 ft
- Depth of ground surface below top of pile = 0.5000 ft

Pile diameters used for p-y curve computations are defined using 2 points.

p-y curves are computed using pile diameter values interpolated with depth over the length of the pile. A summary of values of pile diameter vs. depth follows.

Point No.	Depth Below Pile Head feet	Pile Diameter inches
1	0.000	96.0000
2	27.500	96.0000

Input Structural Properties for Pile Sections:

Pile Section No. 1:

- Section 1 is a round drilled shaft, bored pile, or CIDH pile
- Length of section = 27.500000 ft
- Shaft Diameter = 96.000000 in
- Shear capacity of section = 0.0000 lbs

Ground Slope and Pile Batter Angles

- Ground Slope Angle = 0.000 degrees
- = 0.000 radians
- Pile Batter Angle = 0.000 degrees
- = 0.000 radians

Soil and Rock Layering Information

The soil profile is modelled using 10 layers

Layer 1 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer	=	0.500000	ft
Distance from top of pile to bottom of layer	=	2.500000	ft
Effective unit weight at top of layer	=	115.000000	pcf
Effective unit weight at bottom of layer	=	115.000000	pcf
Friction angle at top of layer	=	30.000000	deg.
Friction angle at bottom of layer	=	30.000000	deg.
Subgrade k at top of layer	=	90.000000	pci
Subgrade k at bottom of layer	=	90.000000	pci

Layer 2 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer	=	2.500000	ft
Distance from top of pile to bottom of layer	=	4.500000	ft
Effective unit weight at top of layer	=	120.000000	pcf
Effective unit weight at bottom of layer	=	120.000000	pcf
Friction angle at top of layer	=	36.000000	deg.
Friction angle at bottom of layer	=	36.000000	deg.
Subgrade k at top of layer	=	225.000000	pci
Subgrade k at bottom of layer	=	225.000000	pci

Layer 3 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer	=	4.500000	ft
Distance from top of pile to bottom of layer	=	6.500000	ft
Effective unit weight at top of layer	=	120.000000	pcf
Effective unit weight at bottom of layer	=	120.000000	pcf
Friction angle at top of layer	=	35.000000	deg.
Friction angle at bottom of layer	=	35.000000	deg.
Subgrade k at top of layer	=	90.000000	pci
Subgrade k at bottom of layer	=	90.000000	pci

Layer 4 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer	=	6.500000	ft
Distance from top of pile to bottom of layer	=	8.500000	ft
Effective unit weight at top of layer	=	115.000000	pcf
Effective unit weight at bottom of layer	=	115.000000	pcf
Friction angle at top of layer	=	30.000000	deg.
Friction angle at bottom of layer	=	30.000000	deg.
Subgrade k at top of layer	=	90.000000	pci
Subgrade k at bottom of layer	=	90.000000	pci

Layer 5 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer	=	8.500000	ft
Distance from top of pile to bottom of layer	=	13.500000	ft
Effective unit weight at top of layer	=	115.000000	pcf
Effective unit weight at bottom of layer	=	115.000000	pcf
Friction angle at top of layer	=	31.000000	deg.
Friction angle at bottom of layer	=	31.000000	deg.
Subgrade k at top of layer	=	90.000000	pci
Subgrade k at bottom of layer	=	90.000000	pci

Layer 6 is stiff clay without free water

Distance from top of pile to top of layer	=	13.500000	ft
Distance from top of pile to bottom of layer	=	18.500000	ft
Effective unit weight at top of layer	=	110.000000	pcf
Effective unit weight at bottom of layer	=	110.000000	pcf
Undrained cohesion at top of layer	=	500.000000	psf
Undrained cohesion at bottom of layer	=	500.000000	psf
Epsilon-50 at top of layer	=	0.020000	
Epsilon-50 at bottom of layer	=	0.020000	

Layer 7 is soft clay, p-y criteria by Matlock, 1970

Distance from top of pile to top of layer	=	18.500000	ft
Distance from top of pile to bottom of layer	=	23.500000	ft
Effective unit weight at top of layer	=	110.000000	pcf

Effective unit weight at bottom of layer = 110.000000 pcf
 Undrained cohesion at top of layer = 400.000000 psf
 Undrained cohesion at bottom of layer = 400.000000 psf
 Epsilon-50 at top of layer = 0.020000
 Epsilon-50 at bottom of layer = 0.020000

Layer 8 is sand, p-y criteria by Reese et al., 1974

Distance from top of pile to top of layer = 23.500000 ft
 Distance from top of pile to bottom of layer = 25.500000 ft
 Effective unit weight at top of layer = 115.000000 pcf
 Effective unit weight at bottom of layer = 115.000000 pcf
 Friction angle at top of layer = 30.000000 deg.
 Friction angle at bottom of layer = 30.000000 deg.
 Subgrade k at top of layer = 90.000000 pci
 Subgrade k at bottom of layer = 90.000000 pci

Layer 9 is stiff clay without free water

Distance from top of pile to top of layer = 25.500000 ft
 Distance from top of pile to bottom of layer = 27.500000 ft
 Effective unit weight at top of layer = 130.000000 pcf
 Effective unit weight at bottom of layer = 130.000000 pcf
 Undrained cohesion at top of layer = 4000. psf
 Undrained cohesion at bottom of layer = 4000. psf
 Epsilon-50 at top of layer = 0.005000
 Epsilon-50 at bottom of layer = 0.005000

Layer 10 is stiff clay without free water

Distance from top of pile to top of layer = 27.500000 ft
 Distance from top of pile to bottom of layer = 38.500000 ft
 Effective unit weight at top of layer = 67.600000 pcf
 Effective unit weight at bottom of layer = 67.600000 pcf
 Undrained cohesion at top of layer = 4000. psf
 Undrained cohesion at bottom of layer = 4000. psf
 Epsilon-50 at top of layer = 0.005000
 Epsilon-50 at bottom of layer = 0.005000

(Depth of the lowest soil layer extends 11.000 ft below the pile tip)

Summary of Input Soil Properties

Layer Num.	Soil Type Name (p-y Curve Type)	Layer Depth ft	Effective Unit Wt. pcf	Cohesion psf	Angle of Friction deg.	E50 or krm	kpy pci
1	Sand	0.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	2.5000	115.0000	--	30.0000	--	90.0000
2	Sand	2.5000	120.0000	--	36.0000	--	225.0000
	(Reese, et al.)	4.5000	120.0000	--	36.0000	--	225.0000
3	Sand	4.5000	120.0000	--	35.0000	--	90.0000
	(Reese, et al.)	6.5000	120.0000	--	35.0000	--	90.0000
4	Sand	6.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	8.5000	115.0000	--	30.0000	--	90.0000
5	Sand	8.5000	115.0000	--	31.0000	--	90.0000
	(Reese, et al.)	13.5000	115.0000	--	31.0000	--	90.0000
6	Stiff Clay	13.5000	110.0000	500.0000	--	0.02000	--

	w/o Free Water	18.5000	110.0000	500.0000	--	0.02000	--
7	Soft	18.5000	110.0000	400.0000	--	0.02000	--
	Clay	23.5000	110.0000	400.0000	--	0.02000	--
8	Sand	23.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	25.5000	115.0000	--	30.0000	--	90.0000
9	Stiff Clay	25.5000	130.0000	4000.	--	0.00500	--
	w/o Free Water	27.5000	130.0000	4000.	--	0.00500	--
10	Stiff Clay	27.5000	67.6000	4000.	--	0.00500	--
	w/o Free Water	38.5000	67.6000	4000.	--	0.00500	--

Static Loading Type

Static loading criteria were used when computing p-y curves for all analyses.

Pile-head Loading and Pile-head Fixity Conditions

Number of loads specified = 2

Load Analysis No.	Load Type	Condition 1	Condition 2	Axial Thrust Force, lbs	Compute Top y vs. Pile Length	Run
1	1	V = 64747. lbs	M = 102862560. in-lbs	93587.	No	
Yes						
2	1	V = 13220. lbs	M = 20877480. in-lbs	58210.	No	
Yes						

V = shear force applied normal to pile axis
M = bending moment applied to pile head
y = lateral deflection normal to pile axis
S = pile slope relative to original pile batter angle
R = rotational stiffness applied to pile head
Values of top y vs. pile lengths can be computed only for load types with specified shear loading (Load Types 1, 2, and 3).
Thrust force is assumed to be acting axially for all pile batter angles.

Computations of Nominal Moment Capacity and Nonlinear Bending Stiffness

Axial thrust force values were determined from pile-head loading conditions

Number of Pile Sections Analyzed = 1

Pile Section No. 1:

Dimensions and Properties of Drilled Shaft (Bored Pile):

Length of Section	=	27.500000 ft
Shaft Diameter	=	96.000000 in
Concrete Cover Thickness (to edge of long. rebar)	=	3.625000 in
Number of Reinforcing Bars	=	52 bars
Yield Stress of Reinforcing Bars	=	60000. psi
Modulus of Elasticity of Reinforcing Bars	=	29000000. psi
Gross Area of Shaft	=	7238. sq. in.
Total Area of Reinforcing Steel	=	40.840704 sq. in.
Area Ratio of Steel Reinforcement	=	0.56 percent
Edge-to-Edge Bar Spacing	=	4.298213 in
Maximum Concrete Aggregate Size	=	0.750000 in
Ratio of Bar Spacing to Aggregate Size	=	5.73

Offset of Center of Rebar Cage from Center of Pile = 0.0000 in

Axial Structural Capacities:

Nom. Axial Structural Capacity = $0.85 F_c A_c + F_y A_s$ = 29980.454 kips
Tensile Load for Cracking of Concrete = -3305.543 kips
Nominal Axial Tensile Capacity = -2450.442 kips

Reinforcing Bar Dimensions and Positions Used in Computations:

Bar Number	Bar Diam. inches	Bar Area sq. in.	X inches	Y inches
1	1.000000	0.785398	43.875000	0.000000
2	1.000000	0.785398	43.555102	5.288547
3	1.000000	0.785398	42.600072	10.499975
4	1.000000	0.785398	41.023838	15.558289
5	1.000000	0.785398	38.849383	20.389729
6	1.000000	0.785398	36.108417	24.923841
7	1.000000	0.785398	32.840909	29.094507
8	1.000000	0.785398	29.094507	32.840909
9	1.000000	0.785398	24.923841	36.108417
10	1.000000	0.785398	20.389729	38.849383
11	1.000000	0.785398	15.558289	41.023838
12	1.000000	0.785398	10.499975	42.600072
13	1.000000	0.785398	5.288547	43.555102
14	1.000000	0.785398	0.000000	43.875000
15	1.000000	0.785398	-5.288547	43.555102
16	1.000000	0.785398	-10.499975	42.600072
17	1.000000	0.785398	-15.558289	41.023838
18	1.000000	0.785398	-20.389729	38.849383
19	1.000000	0.785398	-24.923841	36.108417
20	1.000000	0.785398	-29.094507	32.840909
21	1.000000	0.785398	-32.840909	29.094507
22	1.000000	0.785398	-36.108417	24.923841
23	1.000000	0.785398	-38.849383	20.389729
24	1.000000	0.785398	-41.023838	15.558289
25	1.000000	0.785398	-42.600072	10.499975
26	1.000000	0.785398	-43.555102	5.288547
27	1.000000	0.785398	-43.875000	0.000000
28	1.000000	0.785398	-43.555102	-5.288547
29	1.000000	0.785398	-42.600072	-10.499975
30	1.000000	0.785398	-41.023838	-15.558289
31	1.000000	0.785398	-38.849383	-20.389729
32	1.000000	0.785398	-36.108417	-24.923841
33	1.000000	0.785398	-32.840909	-29.094507
34	1.000000	0.785398	-29.094507	-32.840909
35	1.000000	0.785398	-24.923841	-36.108417
36	1.000000	0.785398	-20.389729	-38.849383
37	1.000000	0.785398	-15.558289	-41.023838
38	1.000000	0.785398	-10.499975	-42.600072
39	1.000000	0.785398	-5.288547	-43.555102
40	1.000000	0.785398	0.000000	-43.875000
41	1.000000	0.785398	5.288547	-43.555102
42	1.000000	0.785398	10.499975	-42.600072
43	1.000000	0.785398	15.558289	-41.023838
44	1.000000	0.785398	20.389729	-38.849383
45	1.000000	0.785398	24.923841	-36.108417
46	1.000000	0.785398	29.094507	-32.840909
47	1.000000	0.785398	32.840909	-29.094507
48	1.000000	0.785398	36.108417	-24.923841
49	1.000000	0.785398	38.849383	-20.389729
50	1.000000	0.785398	41.023838	-15.558289
51	1.000000	0.785398	42.600072	-10.499975
52	1.000000	0.785398	43.555102	-5.288547

NOTE: The positions of the above rebars were computed by LPILE

Minimum spacing between any two bars not equal to zero = 4.298 inches
between bars 48 and 49.

Ratio of bar spacing to maximum aggregate size = 5.73

Concrete Properties:

Compressive Strength of Concrete = 4500. psi

Modulus of Elasticity of Concrete	=	3823676. psi
Modulus of Rupture of Concrete	=	-503.115295 psi
Compression Strain at Peak Stress	=	0.002001
Tensile Strain at Fracture of Concrete	=	-0.0001152
Maximum Coarse Aggregate Size	=	0.750000 in

Number of Axial Thrust Force Values Determined from Pile-head Loadings = 2

Number	Axial Thrust Force kips
1	58.210
2	93.587

Summary of Results for Nominal Moment Capacity for Section 1

Moment values interpolated at maximum compressive strain = 0.003
or maximum developed moment if pile fails at smaller strains.

Load No.	Axial Thrust kips	Nominal Mom. Cap. in-kip	Max. Comp. Strain
1	58.210	104151.110	0.00300000
2	93.587	105468.258	0.00300000

Note that the values of moment capacity in the table above are not factored by a strength reduction factor (phi-factor).

In ACI 318, the value of the strength reduction factor depends on whether the transverse reinforcing steel bars are tied hoops (0.65) or spirals (0.75).

The above values should be multiplied by the appropriate strength reduction factor to compute ultimate moment capacity according to ACI 318, or the value required by the design standard being followed.

The following table presents factored moment capacities and corresponding bending stiffnesses computed for common resistance factor values used for reinforced concrete sections.

Axial Load No.	Resist. Factor	Nominal Ax. Thrust kips	Nominal Moment Cap in-kips	Ult. (Fac) Ax. Thrust kips	Ult. (Fac) Moment Cap in-kips	Bend. Stiff. at Ult Mom kip-in ²
1	0.65	58.210000	104151.	37.836500	67698.	2.5468E+09
2	0.65	93.586667	105468.	60.831333	68554.	2.5844E+09
1	0.75	58.210000	104151.	43.657500	78113.	2.4547E+09
2	0.75	93.586667	105468.	70.190000	79101.	2.4914E+09
1	0.90	58.210000	104151.	52.389000	93736.	1.5681E+09
2	0.90	93.586667	105468.	84.228000	94921.	1.5948E+09

Layering Correction Equivalent Depths of Soil & Rock Layers

Layer No.	Top of Layer Below Pile Head ft	Equivalent Top Depth Below Grnd Surf ft	Same Layer Type As Layer Above	Layer is Rock or is Below Rock Layer	F0 Integral for Layer lbs	F1 Integral for Layer lbs
1	0.5000	0.00	N.A.	No	0.00	14957.
2	2.5000	1.7107	Yes	No	14957.	61479.
3	4.5000	3.7952	Yes	No	76436.	111679.
4	6.5000	6.6747	Yes	No	188115.	137011.
5	8.5000	8.4400	Yes	No	325126.	527601.
6	13.5000	30.4533	No	No	852727.	179940.
7	18.5000	41.5821	No	No	1032667.	144060.
8	23.5000	16.5378	No	No	1176727.	298662.
9	25.5000	12.8613	No	No	1475389.	272823.
10	27.5000	27.0000	No	No	1748212.	N.A.

Notes: The F0 integral of Layer n+1 equals the sum of the F0 and F1 integrals for Layer n. Layering correction equivalent depths are computed only

for soil types with both shallow-depth and deep-depth expressions for peak lateral load transfer. These soil types are soft and stiff clays, non-liquefied sands, and cemented c-phi soil.

Summary of Pile-head Responses for Conventional Analyses

Definitions of Pile-head Loading Conditions:

Load Type 1: Load 1 = Shear, V, lbs, and Load 2 = Moment, M, in-lbs
 Load Type 2: Load 1 = Shear, V, lbs, and Load 2 = Slope, S, radians
 Load Type 3: Load 1 = Shear, V, lbs, and Load 2 = Rot. Stiffness, R, in-lbs/rad.
 Load Type 4: Load 1 = Top Deflection, y, inches, and Load 2 = Moment, M, in-lbs
 Load Type 5: Load 1 = Top Deflection, y, inches, and Load 2 = Slope, S, radians

Load Case No.	Load Type 1	Pile-head Load 1	Load Type 2	Pile-head Load 2	Axial Loading lbs	Pile-head Deflection inches	Pile-head Rotation radians	Max Shear in Pile lbs	Max Moment in Pile in-lbs
1	V, lb	64747.	M, in-lb	1.03E+08	93587.	6.5793	-0.05242	-588166.	1.06E+08
2	V, lb	13220.	M, in-lb	2.09E+07	58210.	0.1790	-9.27E-04	-112579.	2.13E+07

Maximum pile-head deflection = 6.5792756346 inches
 Maximum pile-head rotation = -0.0524226779 radians = -3.003598 deg.

The analysis ended normally.

IBC 1807.3.2.1

Moment (ft·k)	6,428.91	
Shear (k)	48.56	
Caisson diameter (ft)	8	
Caisson height above ground (ft)	0.5	
Caisson height below ground (ft)	27	
Lateral soil pressure (lb/ft ²)	383.33	
Ground to application of force, h (ft)	132.89	
Applied lateral force, P (lb)	48,560	
Lateral soil bearing pressure, S ₁ (lb/ft)	3,450.00	
Diameter, b (ft)	8	
A	4.12	$= (2.34P)/(S_1 b)$
Minimum depth of embedment, d (ft)	26.57	$= 0.5A[1 + (1 + (4.36h / A))^{1/2}]$